

TABLE I
Application Scenarios

Sectors	Industrial	Professional service	Domestic service	Security	Space
Robotic workers	Large structure manufacturing (incl. civil engineering)	Human-like assembly robot			Orbital robot agent
	Robot with integrated process control	Professional cleaning robot			Planetary robot agent
	Rapidly adaptable manufacturing cell	Maintenance robot			
	Postproduction automation (recycling, remanufacturing)	Coordinated mobile manipulators			
	Forestry and agriculture robot				
	Robot automation for small-scale manufacturing				
Robotic co-workers	Mining robot				
	Micromanufacturing robot				
Robotic co-workers	Robotic assistance in industrial environments	Robotic assistance for professionals	Personal robot	Robotic assistance in security contexts	Orbital robot assistant
		Surgical robot	Robotic assistance for physically challenged		Planetary robot assistant
Logistics Robot	Autonomous transport of goods				
	Autonomous transport of people				
Robots for surveillance and intervention		Site protection (domestic and professional)		Border surveillance	
				Security checks of goods and people	
Robots for exploration and inspection	Disaster management			Inspection in environments inaccessible to humans	Orbital robot explorer
	Rehabilitation robot	Underwater robot			Planetary robot explorer
Education and entertainment robots		Motion simulator	Robot companion		
		Robot guide	Robot toy		
		Robot teacher			
		Robot trainer			

is strongly influenced by the investors involved. Designing an underwater system is an expensive task, but results in a cost-effective tool. In the commercial sector of the hydrographic industry, the unmanned underwater vehicle (UUV) is already making a significant impact. Remotely operated vehicles (ROVs) and autonomous underwater vehicle (AUV) are currently routinely deployed for intervention on immersed structures or surveying wide-range areas.

Considerable research effort has been made to improve process performance and control strategies for the various underwater processes over the last half century. However, there are still many problems to overcome. The major efforts on research and development should be focussed on the following topics:

1. Modelling accuracy should be quantified
2. State estimation error has to be bounded
3. Guidance and control systems should exhibit global practical convergence
4. Mission control system and the hardware and software architectures need to be reactive and determinist
5. Efficient vertical communication
6. Minimal horizontal communication
7. Coordinated maneuvers
8. Obstacle avoidance
9. An efficient management of the terrain knowledge

Futuristic robots may be coming soon to an ocean near you. Underwater robotics promises to open a new chapter in the notoriously challenging exploration of earth's ocean. Future

scope of ocean exploration due to underwater robotics includes:

1. Intelligence, surveillance and reconnaissance
2. Mine countermeasures
3. Anti-submarine warfare
4. Inspection/identification
5. Oceanography
6. Communication/navigation network nodes
7. Payload delivery
8. Information operations
9. Time-critical strike
10. Underwater welding

Nanorobotics

Nanorobotics is the emerging technology field creating machines or robots whose components are at or close to the scale of a nanometer. More specifically, nanorobotics refers to the nanotechnol-